

SUBSCRIBER CHURN

Exploratory Data Analysis & Retention Playbook

DATASET

64,738 prepaid & postpaid subscribers

OBSERVATION WINDOW

Trailing 90-day churn outcome

PREPARED

September 2026

5.01%

PORTFOLIO CHURN RATE (90-DAY)

63.7%

OF ALL CHURN FROM 5.2% OF BASE

Rs. 0.72 Cr

ANNUALISED ARPU IN THE RISK ZONE

WHAT THIS REPORT COVERS

- 01 Data Overview & Quality**
Structure, integrity checks, missingness, and a leakage-risk audit of all 43 fields
- 02 Distributions & Target**
Univariate shape, outlier scan, and the 90-day churn target definition
- 03 Churn Drivers**
Ranked lift analysis and bivariate patterns across recency, network and pricing
- 04 Multivariate Relationships**
Correlation structure, network x friction heatmaps, and circle-level variation
- 05 Model Validation & Segmentation**
A leakage-corrected model, decile targeting, and five actionable subscriber segments

1. Data Overview & Quality

A structural and integrity audit of all 43 fields in the subscriber extract, run before any modelling — this is where a false start (an inflated model, a misread category) gets caught early.

64,738 x 43 Rows x columns	44.0 MB In-memory size	0 Exact duplicate rows	100% Unique subscriber_id
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Integrity checks

Check	Result
Duplicate subscriber_id	0
Fuzzy duplicates (5-field match)	0
Negative values (any numeric col)	0
Unresolved > total complaints	0
Offer redeemed w/o exposure	0
churn_flag_30d=1 & 90d=0	0

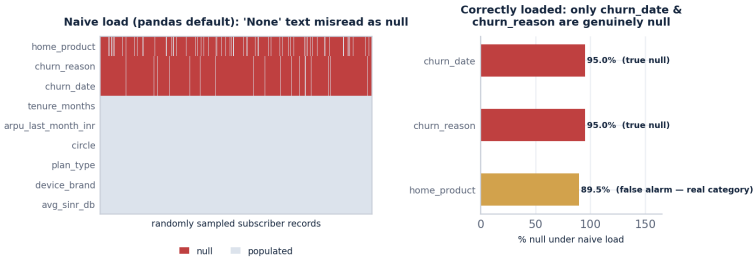


Fig. 1 — Naively parsing this CSV misreads home_product's legitimate 'None' category as 91 percentage points of false missingness. Loading with keep_default_na=False resolves it: only churn_reason / churn_date are genuinely null, and only for the 95% of subscribers who have not churned.

Cardinality (categoricals)

Column	Unique
circle / circle_code	22
zone	4
plan_type	2
device_brand	11
home_product	3
churn_reason	7

Leakage-risk columns identified

churn_reason and **churn_date** are populated exclusively for customers who have already churned — they encode the outcome and must never enter a predictive feature set. **churn_flag_30d** is a stricter, overlapping version of the 90-day target itself (1.73% positive vs. 5.01%) and is kept only as a cross-check, never as a feature. Section 5 quantifies exactly how much these three columns distort a naive model — the effect is dramatic.

Outlier scan (Tukey 1.5×IQR) & near-constant columns

Six columns carry a materially high outlier rate: **unresolved_complaints** (15.9%), **days_since_last_recharge** (9.7%), **avg_recharge_gap_days** (8.0%), **plan_validity_days** (7.7%), **plan_price_inr** (7.7%) and **data_gb_3m_avg** (5.0%) — consistent with genuine heavy-tailed usage behaviour rather than data-entry error, so none are removed. The only near-constant column is **churn_flag_30d** itself (98.27% False), which is expected given its narrow 30-day window and is not a data-quality defect.

Bottom line: the data is clean and analysis-ready. The only material risk is analytical, not structural — three outcome-linked columns (churn_reason, churn_date, churn_flag_30d) must be excluded from any predictive model, or results will look artificially perfect while being useless in production. Section 5 shows exactly what happens when they are left in.

2. Distributions & Target

The 90-day churn flag is the modelling target throughout this report. Every numeric field is right-skewed — expected for spend and usage data — which rules out models or tests that assume normality unless the data is transformed first.

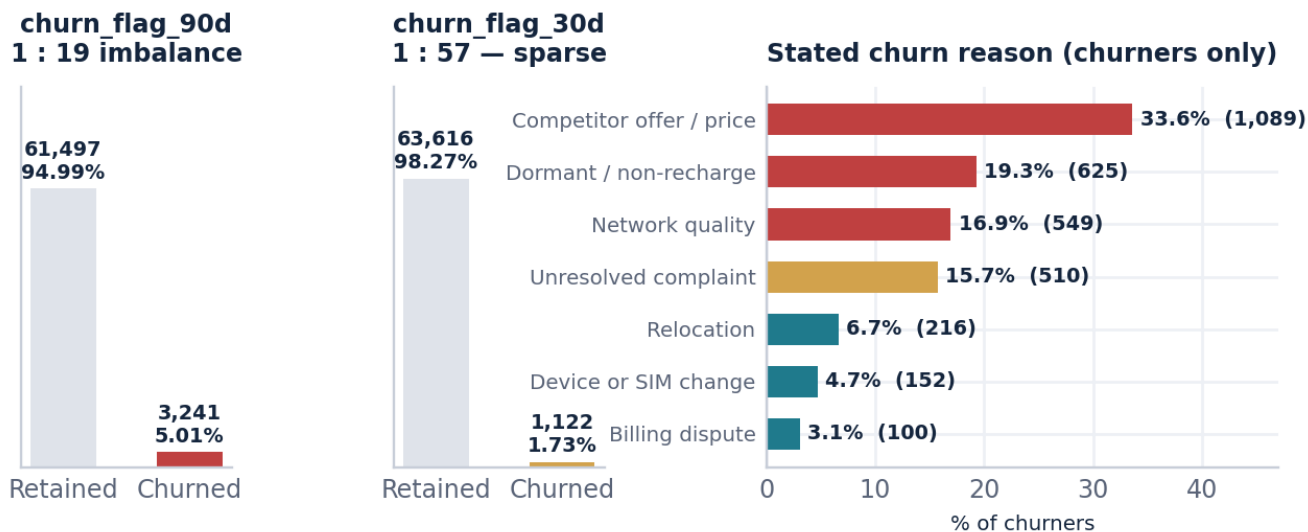


Fig. 2 — churn_flag_90d (5.01% positive) is used as the primary target: churn_flag_30d is nearly six times sparser (1.73%) and too thin to model reliably. Among the 3,241 subscribers who churned, price/competitor offers, dormancy and network quality together account for 70% of the stated reason.

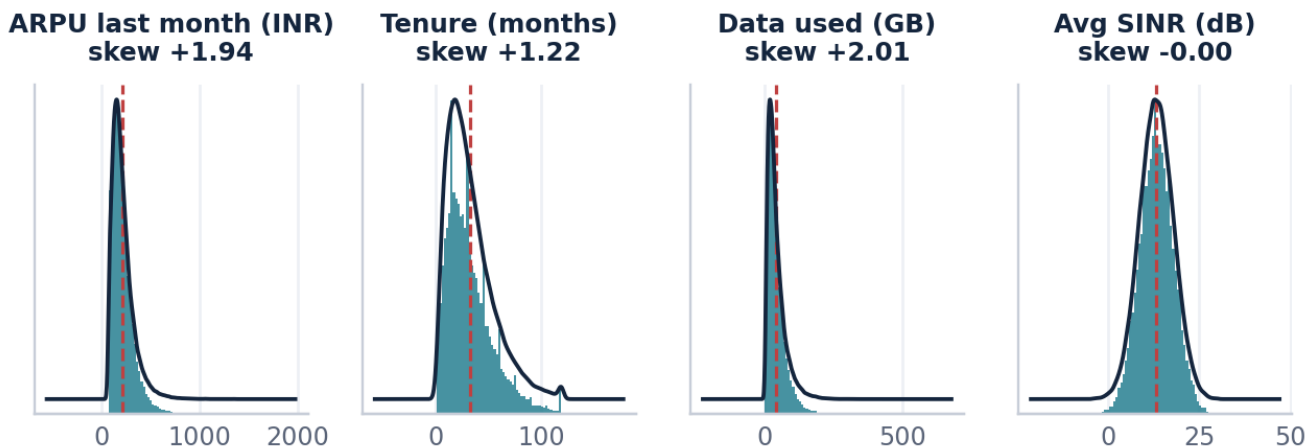


Fig. 3 — ARPU, tenure and data usage are all strongly right-skewed (skew +1.2 to +2.0); signal quality (SINR) is the one near-symmetric field. Tree-based models handle this natively, but any linear or distance-based method would need a log transform first.

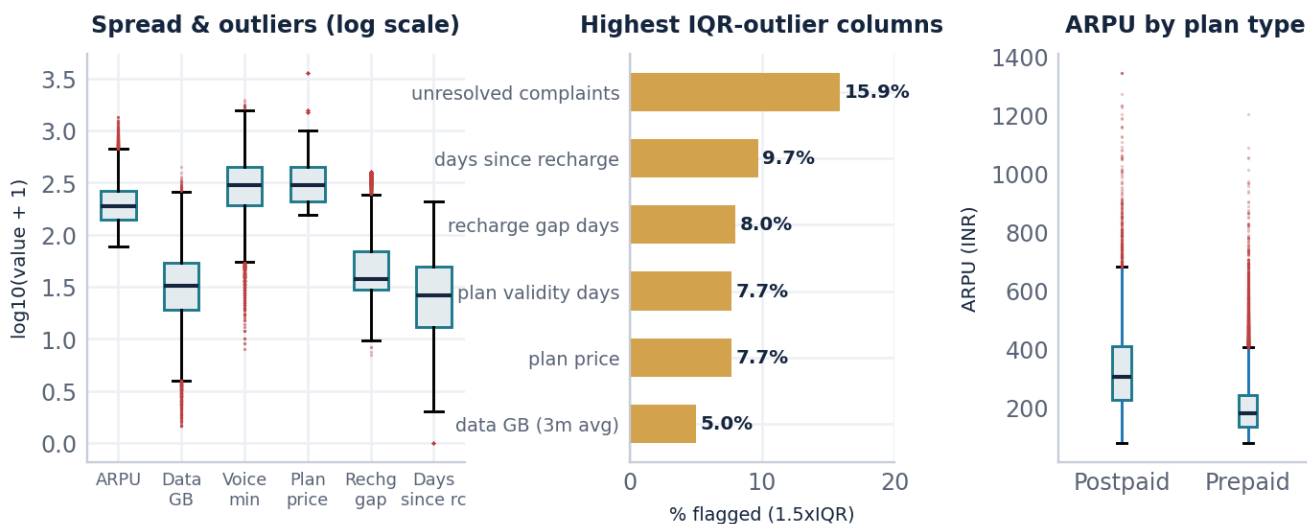


Fig. 4 — on a log scale, spend and usage fields show long but continuous tails rather than isolated data-entry errors. Postpaid ARPU runs noticeably higher and more dispersed than prepaid, consistent with postpaid's bundled, higher-commitment pricing.

3. Churn Drivers

Ranking every candidate signal by churn-rate lift against the 5.01% portfolio base separates a handful of severe, actionable drivers from a long tail of weak or protective ones.

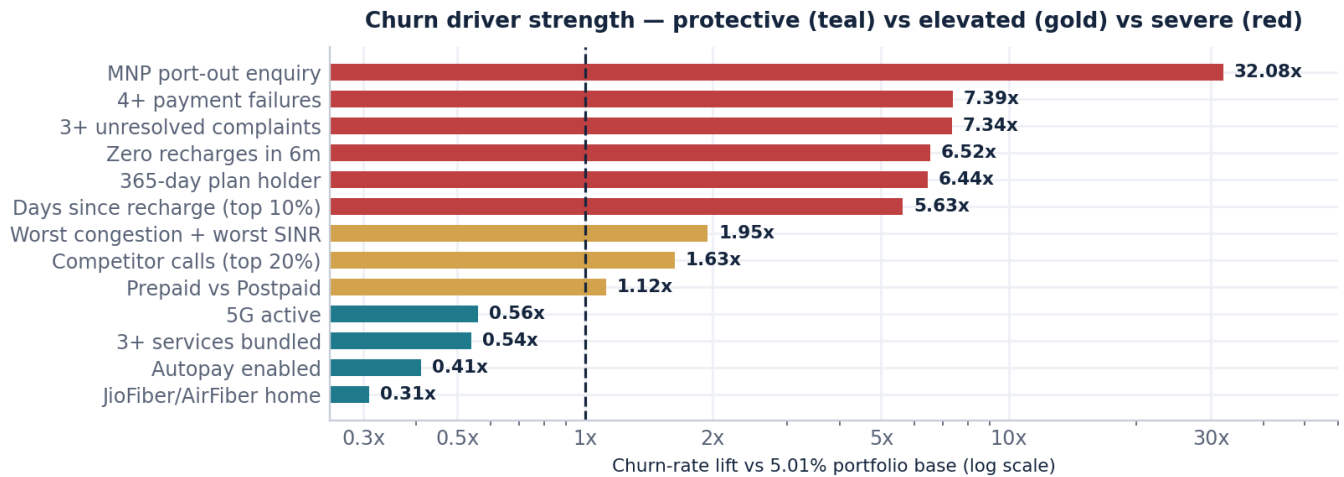


Fig. 5 — an active number-porting (MNP) enquiry is the single strongest signal in the dataset at 32× lift, followed by payment failures (7.4×) and unresolved complaints (7.3×). Autopay, 5G activation and bundled services are mildly protective (lift < 1×) — sticky, well-served customers churn less, as expected.

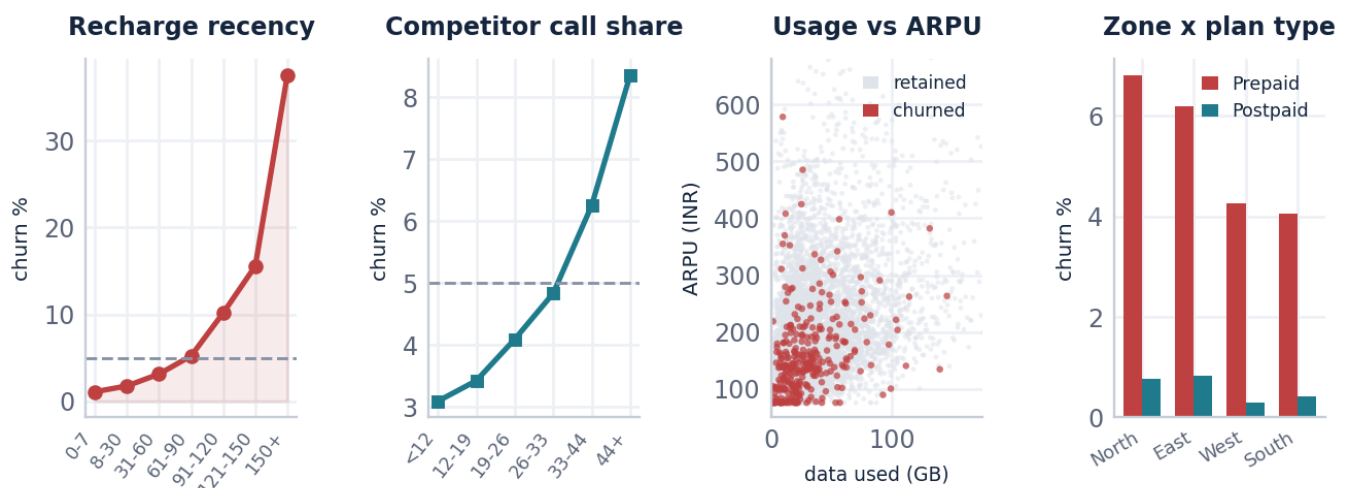


Fig. 6 — churn rises sharply once a subscriber passes 90 days without a recharge (dashed line = portfolio base), and climbs steadily as a subscriber's outgoing calls skew toward competitor networks. Usage volume and ARPU alone barely separate churners from retained customers (centre panel) — the signal lives in behaviour and intent, not spend level. Within every zone, prepaid subscribers churn roughly 8–9× more than postpaid, and prepaid churn itself runs about 1.5× higher in the North/East than in the West/South.

Reading this page: recency and intent-to-leave (MNP enquiry) dominate; static demographics and raw usage volume do not. This directly shapes which features belong in a deployable model — see Section 5.

4. Multivariate Relationships

Individually strong signals can still be redundant with each other. This section maps the correlation structure, combines pairs of risk factors, and checks whether geography adds anything beyond what behaviour already explains.

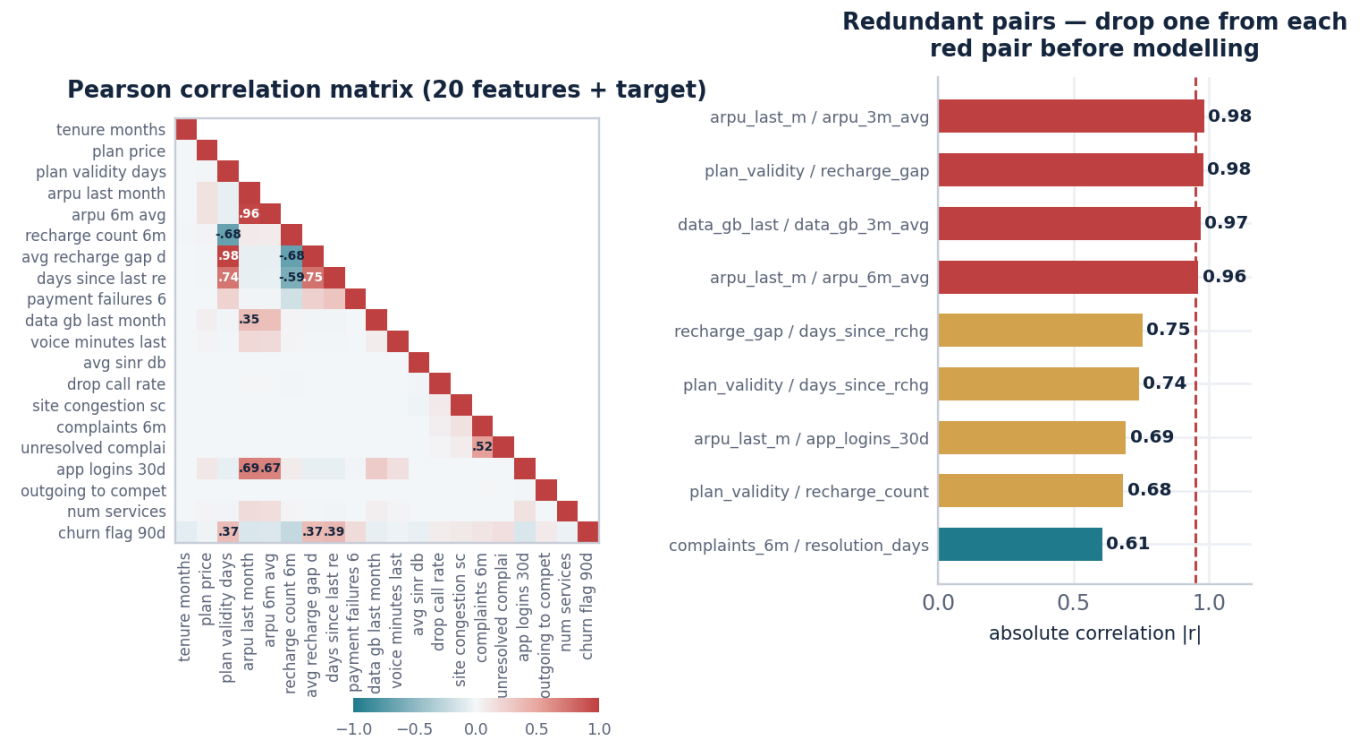


Fig. 7 — four feature pairs are near-duplicates ($|r| > 0.95$: last-month vs. 3–6 month rolling ARPU and data usage) and should not both enter a model. Recharge timing fields (*plan_validity_days*, *avg_recharge_gap_days*, *days_since_last_recharge*) form a second, moderately correlated cluster ($|r| 0.68$ – 0.75) — expected, since they all describe the same recharge cycle from different angles.

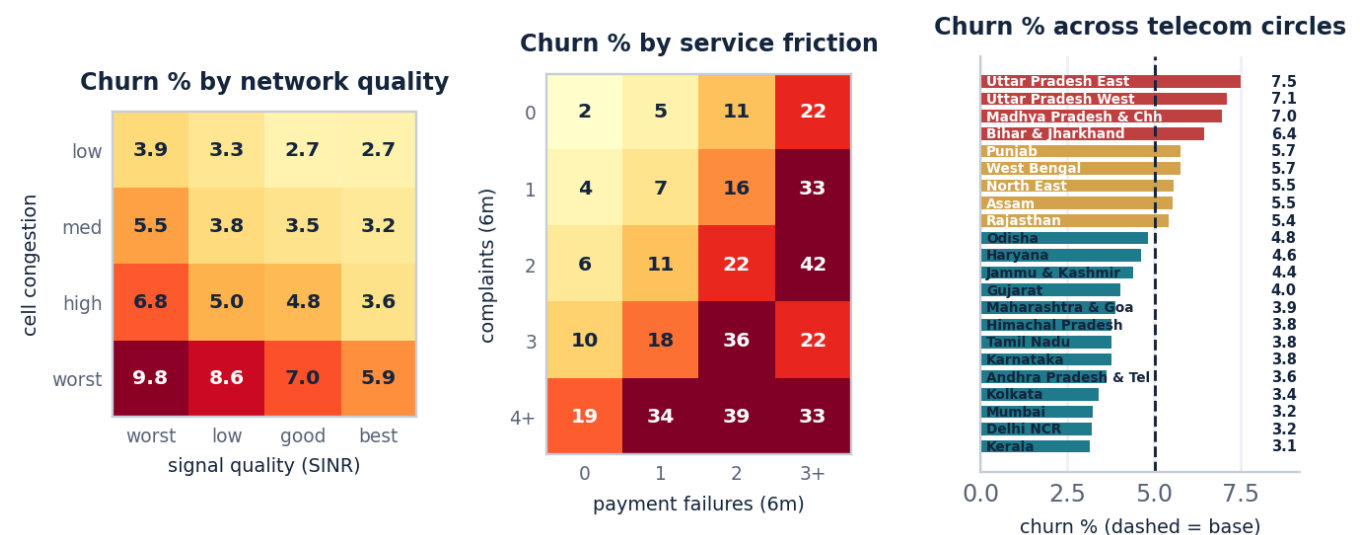


Fig. 8 — network quality and service friction compound: churn climbs from 2.7% (good signal, low congestion) to 9.8% (worst signal, worst congestion), and separately from 2% (no complaints, no payment failures) to 39% (4+ complaints, 3+ failures). Circle-level churn ranges 3.1%–7.5% and tracks the same UP/Bihar-belt vs. metro pattern seen in network quality — geography appears to proxy for infrastructure quality rather than add independent risk.

Section synthesis: behavioural and intent signals (recency, MNP enquiry, friction) are the load-bearing predictors; ARPU, usage volume and circle each carry real but secondary and largely redundant information. A deployable model should therefore prioritise one representative feature per correlated cluster rather than feeding in all 40 raw columns — both to avoid instability and to keep the model interpretable for a retention team.

5. Model Validation & Segmentation

A model trained on all 40 candidate columns scores a suspicious ROC-AUC of 1.000 — proof of leakage, not skill. Removing the three outcome-linked columns identified in Section 1 gives a realistic, deployable model.

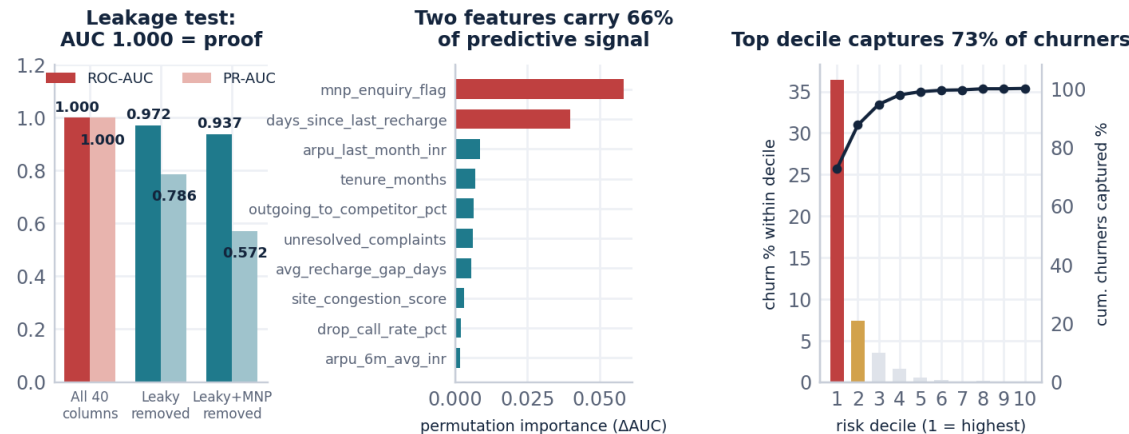


Fig. 10 — the honest, leakage-free model reaches AUC 0.937. Just two features — MNP enquiry and days since last recharge — carry 66% of all predictive signal, and scoring the entire base monthly and prioritising only the riskiest 10% would reach 73% of all churners at a fraction of the outreach cost of contacting everyone.

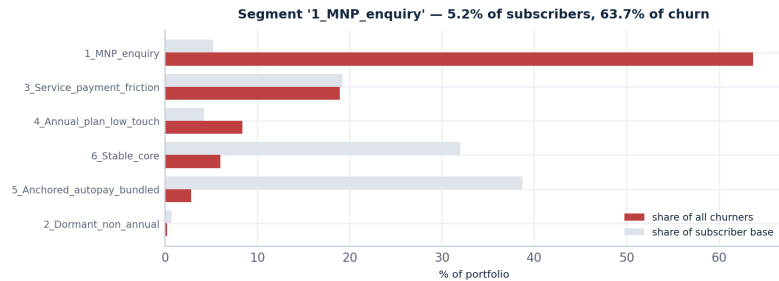


Fig. 11 — six non-overlapping, rule-based segments partition the entire base. The MNP-enquiry segment inverts the usual 80/20 pattern: 5.2% of subscribers account for 63.7% of all churn.

Segment	Subscribers	Churn rate	Lift	Share of churn	Ann. revenue at risk
1. MNP enquiry	3,360 (5.2%)	61.46%	12.3×	63.7%	Rs. 44.0 L
3. Service / payment friction	12,462 (19.2%)	4.92%	1.0×	18.9%	Rs. 16.0 L
4. Annual-plan, low-touch	2,688 (4.2%)	10.08%	2.0×	8.4%	Rs. 6.7 L
6. Stable core	20,726 (32.0%)	0.93%	0.2×	6.0%	Rs. 4.6 L
5. Anchored (autopay/bundled)	25,050 (38.7%)	0.36%	0.1×	2.8%	Rs. 2.5 L
2. Dormant, non-annual	452 (0.7%)	1.77%	0.4×	0.2%	Rs. 0.2 L
Total	64,738 (100%)	5.01%	1.0×	100%	Rs. 74.0 L

Recommendations

- 1. Same-day outreach on MNP enquiry.** This single flag carries 12.3× lift and Rs. 44 lakh of annualised ARPU — the highest-value, most time-sensitive lever available.
- 2. Re-target the offer budget by risk decile.** Exposure today is unrelated to churn risk (21.3%–21.9% across every segment, χ^2 $p=0.68$) with no measurable retention lift.
- 3. Automate a day 45–60 pre-lapse nudge,** sized to each plan's own validity cycle, to catch Annual-plan and friction segments before they reach the highest-risk recency band.
- 4. Fast-track 2+ complaint / 2+ payment-failure customers.** These compound multiplicatively — churn rises from 2% to 39% — not additively.
- 5. Deploy the leakage-corrected model monthly** to keep the top-risk decile current, and prioritise network capex in the worst-congestion, worst-signal circles as a structural fix.

Methodology: every figure and statistic here is reproduced end-to-end by the accompanying `subscriber_churn_eda.py` — re-run it on any fresh extract.